AI Language Model Benchmarking on Resource-Constrained Devices: A Performance Analysis

This infographic summarizes a project focused on benchmarking various AI language models (LLMs) on resource-constrained devices, specifically a Raspberry Pi and a MacBook. The goal is to evaluate the efficiency and performance trade-offs of models like TinyLlama, Phi-3 Mini, Mistral, Orca, Gemma 2B, Codellama, and Llama2-7B-Chat. Key metrics analyzed include RAM usage, inference latency, and model size. The infographic highlights the methodology used to ensure fairness, presents a side-by-side comparison of model performance, and identifies the most optimized models for edge computing applications.

AI Model Performance Overview

Project Overview

Goal: Evaluate the performance and efficiency of various LLMs on resource-constrained devices.

Devices:

- Raspberry Pi (Core: Raspberry Pi)
- MacBook (Core: MacBook)

Computing Devices

Models:

- TinyLlama
- Phi-3 Mini
- Mistral
- Orca
- Codellama
- Gemma 2B
- Llama2-7B-Chat

LLMs

Focus Areas:

- Low Resource (Label)
- Quantized Models (Label)
- Edge AI (Label)

AI applications

Methodology: Ensuring Fair Comparisons

Steps:

- **Device Reboot:** Reboot the device before each test to ensure a clean state.
- **Identical Prompts:** Use the same set of prompts for all models.
- **Controlled Environment:** Minimize background processes during testing.
- **Multiple Runs:** Conduct multiple runs and average the results for accuracy.

Benchmarking Cycle for AI Models

Visual: Flowchart illustrating the steps.

Performance Metrics

Metrics Measured:

- **CPU Usage:** Percentage of CPU utilized during inference.
- **RAM Usage:** Amount of RAM consumed by the model.
- **Inference Latency:** Time taken to generate a response (in seconds).
- **Model Size:** Size of the model file (in MB).

Model Performance Metrics

Visual: AI Chip/Neural Network Symbol

Side-by-Side Comparison

[Note: this section would ideally contain bar charts or icon-based comparisons. The following is a textual representation of what the visual would convey.]

CPU Usage:

- TinyLlama: Low
- Phi-3 Mini: Low-Medium
- Mistral: Medium
- LLaVA: Medium
- Orca: Medium-High
- Codellama: Medium
- Gemma 2B: Low-Medium
- Llama2-7B-Chat: High

Model complexity ranges from low to high resource demand.

RAM Usage:

- TinyLlama: Very Low
- Phi-3 Mini: Low
- Mistral: Medium
- LLaVA: High
- Orca: Medium
- Codellama: Medium
- Gemma 2B: Low
- Llama2-7B-Chat: High

Model size spectrum from tiny to large.

Inference Latency:

- TinyLlama: Fast
- Phi-3 Mini: Fast
- Mistral: Medium
- LLaVA: Slow
- Orca: Medium
- Codellama: Medium
- Gemma 2B: Fast
- Llama2-7B-Chat: Slow

Model Size:

- TinyLlama: Smallest
- Phi-3 Mini: Small
- Mistral: Medium
- LLaVA: Large
- Orca: Medium
- Codellama: Medium
- Gemma 2B: Small
- Llama2-7B-Chat: Large

Top Performers in Inference Latency

Model Size Rankings

Visual: Bar charts comparing each metric across all models.

Key Findings & Conclusions

- **TinyLlama:** Performs best under memory constraints, making it ideal for extremely limited environments.
- **Phi-3 Mini:** Offers a balance between speed and accuracy, suitable for general-purpose edge applications.
- **Gemma 2B:** Offers fast inference speeds with low resource consumption.
- **Llama2-7B-Chat & LLaVA:** Require significant resources and are less suitable for edge deployment on constrained devices.
- **Quantization:** Quantization techniques significantly reduce model size and improve inference speed across all models.

Top AI Models for Resource-Constrained Environments

Optimized Models for Edge Computing

Top Recommendations:

1. TinyLlama (For minimal resource usage)
2. Phi-3 Mini (For balanced performance)
3. Gemma 2B (For fast inference)

Top AI Language Models for Resource-Constrained Devices